Healthcare Data Cleaning & Preparation Report

Project: Healthcare Data Cleaning & Audit (`HealthCare_Data_Messy.xlsx`)
Analyst: Sara Sayed
Tool Used: Power Query / Microsoft Excel

1. Project Overview & Objective

This documentation outlines the complete step-by-step data cleaning, normalization, and quality assurance process applied to HealthCare_Data_Messy.xlsx. The primary objective is to transform raw, inconsistent healthcare transactional data into a pristine, structured dataset suitable for advanced analytical reporting, PivotTable aggregation, and executive insights.

2. Global Data Cleaning Strategy

To maintain dataset integrity and avoid biasing downstream analytics, the following overarching rules were established:

- **Record Preservation:** Missing values in categorical fields are imputed with clear fallback labels (e.g., `Unknown Department`, `Unknown Doctor`) rather than dropping entire rows, preventing the loss of financial and clinical metrics.
- **Missing Identifiers:** Records lacking core transaction IDs (`VisitID`) or patient links (`PatientID`) are removed, as unidentifiable records compromise database relational integrity.
- **Statistical Integrity for Numeric Data:** Missing numeric values (such as `Age`) are retained as true `null` (blank) in Excel to avoid distorting measures of central tendency (means, standard deviations).

3. Step-by-Step Column Cleaning & Transformations

Column	Primary Action Executed	Decision & Rationale
VisitID & PatientID	Inspected for missing values (`null`) and structural formatting.	Verified primary key presence. Records lacking valid identifiers were removed to guarantee unique transaction tracking.
PatientName	Applied `Trim` and `Clean` functions to remove whitespace and non-printable characters.	Replaced `null` values with `"Unknown Patient Name"` to keep patient profiles standardized across reports.
Age	Evaluated for impossible values, negative numbers, and extreme outliers (>100 years).	Corrected negative values via absolute conversion (`Scientific.Absolute`). Outliers and unrecorded ages were set to `null` to preserve distribution accuracy.
Gender	Standardized categorical representations (`M`, `F`, `Unknown`).	Cleared inconsistent strings and imputed missing values (`null`) with `"Unknown"`.
Department	Applied `Trim` and `Clean`, removed stray dashes (e.g., `Cardiology-`), and normalized naming.	Replaced missing entries (`null`) with `"Unknown Department"`.
DoctorName, Diagnosis, & Treatment	Cleared hidden spaces and standardized text casing across all clinical text attributes.	Imputed missing values with `"Unknown Doctor"`, `"Unknown Diagnosis"`, and `"Unknown Treatment"` respectively.

Column	Primary Action Executed	Decision & Rationale
PaymentMethod	Normalized inconsistent spellings (`Cash`, `Credit Card`, `Debit Card`, `Insurance`).	Imputed missing values (`null`) with `"Unknown Payment"` to enable accurate financial grouping in revenue breakdowns.
VisitDate & LabTest	Converted `VisitDate` to explicit `Date` type. Inspected `LabTest` categories (`Blood Test`, `CT Scan`, `MRI`, `X-Ray`, `None`).	Preserved explicit `"None"` entries as valid clinical outcomes. Replaced true `null` values with `"Unknown Test"`.

4. Logical Validation & Date Sanity Check

- **Objective:** Ensure chronological logical consistency between initial clinical visits and subsequent follow-ups.
- **Implementation:** Created a conditional audit column (`DateCheck`) in Power Query using the following logic:
  `If FollowUpDate < VisitDate Then "Invalid Date" Else "Valid"`
- **Result: 100% Pass Rate.** Zero records exhibited backwards dates (`FollowUpDate` preceding `VisitDate`).
- **Documentation Note:** A screenshot of the `DateCheck` audit was captured for quality records, and the temporary validation column was safely removed prior to executing `Close & Load`.

5. Final Dataset Status & Excel Table Formatting

- **Target Sheet:** `CleanedData`
- **Structure:** Converted output to a dynamic Excel Table (`Ctrl + T`) to preserve formula integrity, support automatic range expansions, and serve as a reliable source for downstream PivotTables.

- **Conclusion:** The cleaned dataset contains zero orphan records, normalized categorical values, and verified chronological logic. It is fully validated and ready for exploratory data analysis (EDA).

Analytical Questions Plan

Axis / Category	Q_ID	Analytical Question	Description / Objective
1. Operational & Departmental Analysis	Q1	What are the top medical departments by number of visits, and which have the lowest volume?	Evaluate demand and patient distribution across departments.
	Q2	What are the most frequent medical diagnoses and treatments in the hospital?	Identify common conditions and prevailing therapeutic protocols.
	Q3	What is the distribution of lab tests among patients, and what percentage had no tests ordered (None)?	Analyze laboratory and imaging test request rates.
2. Financial & Payment Analysis	Q4	What is the most frequently used payment method among patients?	Understand patient financial behavior and payment preferences.
	Q5	How do payment methods distribute across various medical departments (e.g., reliance on insurance vs. cash)?	Link revenue streams and payment methods to specialized clinics.
3. Physician Performance	Q6	Which physicians have handled the highest patient volume?	Measure clinical workload and staff productivity.

Axis / Category	Q_ID	Analytical Question	Description / Objective
	Q7	What is the distribution of medical cases per physician according to department specialization?	Analyze sub-specialization and case distribution across doctors.
4. Patient Demographics & Time Trends	Q8	What is the overall average age of patients, and how does average age vary by medical department?	Determine high-traffic age brackets per department.
	Q9	How are patients distributed by gender (Male, Female, Unknown) across medical departments?	Study demographic composition of patient traffic.
	Q10	What is the temporal trend of visits over months and years (Time-series analysis)?	Identify peak seasons and operational surges.
5. Advanced Analytics (Efficiency & Compliance)	Q11	What is the average time gap (in days) between the initial visit date and the follow-up date?	Measure clinical scheduling efficiency and compliance.
	Q12	Do specific departments (e.g., Physical Therapy or Orthopedics) record longer follow-up periods than others?	Compare follow-up adherence and timelines across departments.
	Q13	What is the proportion of cases recorded without a follow-up date (Null/Blank) compared to scheduled follow-ups?	Measure patient care journey completion rates.
6. Advanced Analytics	Q14	What is the relationship between patient age and the	Analyze hidden patterns associated with age brackets.

Axis / Category	Q_ID	Analytical Question	Description / Objective
(Multivariate Cross-Analysis)		most requested diagnoses or lab tests?	
	Q15	How do payment methods correlate with specific treatment types or surgical procedures?	Study the financial correlation with clinical interventions.

Section 5: Operational & Departmental Analysis Results

5.1 Q1: Departmental Visit Volume & Share Analysis

- **Objective:** Evaluate patient demand and distribution across medical departments.
- **Data Summary:** Total analyzed visits = 13,000 across 9 specific medical departments plus 1 unassigned category (`Unknown Department`).
- **Key Findings:**
    - **Top Performing Department:** Cardiology leads patient volume with 1,587 visits (12.21% of total visits).
    - **Balanced Departmental Demand:** All 8 specialized departments (Cardiology, Orthopedics, Gynecological care, Oncology, Neurology, Dermatology, ENT, Pediatrics) demonstrate highly balanced demand, ranging between 1,500 and 1,587 visits each (~12% total share per department).
    - **Lowest Volume:** Pediatrics recorded the lowest visits among identified departments (1,500 visits, 11.54%).
    - **Unassigned Records (`Unknown Department`):** Accounting for 650 visits (5.00%), representing records where clinical department affiliation was missing prior to cleaning.
- **Executive & Managerial Recommendations:**
    - Resource and staffing allocation across major specialties should remain evenly distributed given equal patient intake.
    - Cardiology requires primary priority for peak-hour clinical room assignment.
    - Data capture at patient intake must be audited to reduce the 5% unassigned department records.

5.2 Q2: Diagnosis and Treatment Analysis

- **Objective:** Identify the most frequent medical diagnoses and prevailing therapeutic protocols in the hospital.
- **Data Summary:** Total analyzed visits = 13,000 across multiple clinical diagnoses and treatment types.
- **Key Findings (Diagnoses):**
    - **Top Diagnoses:** Joint Pain and Pregnancy lead the diagnostic volume with 430 visits (3.3%) and 424 visits (3.3%) respectively.
    - **Other Frequent Conditions:** Hypertension and Migraine recorded 404 visits each (3.1%), followed closely by Skin Rash (410 visits, 3.2%) and Leukemia (403 visits, 3.1%).
    - **Unassigned Records (`Unknown Diagnosis`):** Accounted for 650 visits (5.0%).
- **Key Findings (Treatments):**
    - **Top Treatments:** Consultation is overwhelmingly the most frequent treatment type with 3,006 visits (23.1%), followed by Medication (2,308 visits, 17.8%) and Surgery (2,397 visits, 18.4%).
    - **Other Treatments:** Ultrasound (418 visits, 3.2%), Radiation (400 visits, 3.1%), and Laser Therapy / Lifestyle Advice (398 visits, 3.1% each).
    - **Unassigned Records (`Unknown Treatment`):** Accounted for 650 visits (5.0%).
- **Executive & Managerial Recommendations:**
    - Given that consultations, medications, and surgeries make up the vast majority of clinical interventions, operational capacity and staffing in outpatient clinics and surgical units must be prioritized.
    - Diagnostic tracking needs enhanced data capture at intake to minimize the 5% unassigned category.

5.3 Q3: Lab Tests and Imaging Distribution Analysis

- **Objective:** Analyze laboratory and imaging test request rates and identify the proportion of cases where no tests were ordered (`None`).
- **Data Summary:** Total analyzed visits = 13,000 across various lab tests, imaging categories, and unassigned records.
- **Key Findings:**
    - **High Test Request Rates:** The distribution across specific tests is remarkably uniform, led by **MRI** with 2,484 visits (19.1%), **X-Ray** with 2,491 visits (19.2%), **Blood Test** with 2,435 visits (18.7%), and **CT Scan** with 2,433 visits (18.7%).

    - **Cases with No Tests Ordered (`None`):** Accounted for 2,507 visits (19.3%), representing patients who received clinical consultations, medications, or therapies without requiring laboratory or radiological testing.
    - **Unassigned Records (`Unknown Test`):** Accounted for 650 visits (5.0%), indicating missing test tracking data prior to cleaning.
- **Executive & Managerial Recommendations:**
    - The high volume of diagnostic and imaging requests (MRI, X-Ray, Blood Test, CT Scan) requires maintaining robust, round-the-clock laboratory and radiology staffing to prevent diagnostic bottlenecks.
    - The substantial proportion of visits with no tests ordered (~19.3%) highlights the prevalence of outpatient consultations and routine checkups that do not require clinical diagnostics.

Section 6: Financial & Payment Analysis Results

6.1 Q4: Payment Method Distribution & Preference Analysis

- **Objective:** Understand patient financial behavior and identify the most frequently used payment methods.
- **Data Summary:** Total analyzed visits = 13,000 across multiple payment categories plus unassigned records.
- **Key Findings:**
    - **Top Payment Method:** Credit Card leads patient transactions with 3,168 visits (24.4% of total visits).
    - **Balanced Financial Distribution:** The remaining primary payment methods show remarkably close, well-balanced distribution: Cash accounts for 3,107 visits (23.9%), Insurance accounts for 3,083 visits (23.7%), and Debit Card accounts for 2,992 visits (23.0%).
    - **Unassigned Records (`Unknown Payment`):** Accounted for 650 visits (5.0%), indicating unrecorded payment attributes prior to data cleaning.
- **Executive & Managerial Recommendations:**
    - Given the high volume of digital and card payments (Credit and Debit cards combined total 47.4%), the hospital's electronic point-of-sale (POS) and online billing infrastructure must be robust and secure.

    - Insurance processing and cash handling desks must maintain parallel operational capacity since Cash (~23.9%) and Insurance (~23.7%) represent substantial, nearly identical shares of patient transactions.
    - Intake personnel must enforce strict logging of payment methods to eliminate the 5% unassigned transactions.

6.2 Q5: Cross-Departmental Payment Distribution Analysis (Department_Payment_Matrix)

- **Objective:** Analyze how payment methods are distributed across various medical departments to identify any specialized reliance on insurance, cash, or credit options.
- **Data Summary:** Total analyzed visits = 13,000 evaluated across a 9×5 department-by-payment matrix.
- **Key Findings:**
    - **Uniform Payment Distribution Across Specialties:** Payment methods demonstrate near-perfect balance across all medical departments without concentration risk in any single specialty.
    - **Departmental Breakdown:** Each major department (Cardiology, Dermatology, ENT, Gynecology, Neurology, Oncology, Orthopedics, Pediatrics) shows payment proportions closely clustered around ~2.8% to 3.0% of total hospital visits for Cash, Credit Card, Debit Card, and Insurance.
    - **Highest Credit Card Adoption:** Orthopedics recorded the highest proportion of Credit Card transactions at 3.2% of total visits (416 visits), followed by Cardiology and Gynecology at 2.9% to 3.0%.
    - **Insurance Reliance:** ENT leads in relative Insurance utilization at 3.0% of total visits, closely followed by Cardiology (2.9%) and Orthopedics (2.8%).
- **Executive & Managerial Recommendations:**
    - Standardize point-of-sale (POS) terminal placement evenly across all outpatient clinics rather than restricting card terminals to specific high-volume departments.
    - Maintain centralized billing operations since no single clinical department relies exclusively on a single payment channel.

Section 7: Physician Performance Results

7.1 Q6: Top Performing Physicians & Workload Distribution Analysis

- **Objective:** Measure clinical workload distribution and identify top-performing physicians by patient volume.
- **Data Audit & Quality Insight:**
    - **High Physician Dispersion:** The dataset contains 670 distinct physician names across 13,000 total visits, resulting in an average of ~19 visits per physician over two years. This highlights data entry variations and formatting inconsistencies.
    - **Unassigned Records (`Unknown Doctor`):** Accounted for 650 visits (5.0%), indicating unrecorded physician tracking prior to data cleaning. Per professional analytical standards, this category was treated as a data quality gap and excluded from the active top-performer ranking.
- **Key Findings (Top 10 Busiest Physicians):**
    - **Michael:** 288 visits (2.2% of total visits)
    - **David:** 197 visits (1.5%)
    - **John:** 193 visits (1.5%)
    - **James:** 186 visits (1.4%)
    - **Jennifer:** 181 visits (1.4%)
    - **Christopher:** 158 visits (1.2%)
    - **Robert:** 141 visits (1.1%)
    - **William:** 135 visits (1.0%)
    - **Jessica:** 134 visits (1.0%)
    - **Lisa:** 128 visits (1.0%)
- **Executive & Managerial Recommendations:**
    - Standardize doctor master data in the EHR System to eliminate typos, duplicate name variations, and unassigned records (resolving the 5% `Unknown Doctor` gap).
    - Implement active staff tracking logs to evaluate clinical productivity accurately based on active operational periods.

7.2 Q7: Cross-Departmental Physician Distribution Analysis (Doctor_Department_Matrix)

- **Objective:** Analyze case and visit distribution per physician across specialized clinical departments.

- **Data Audit & Critical Quality Insight (Anomalous Distribution Pattern):**
    - **Cross-Departmental Physician Duplication:** High-volume physicians (e.g., Michael, John, David) recur as top-performing doctors across almost all specialized departments (Cardiology, Dermatology, ENT, Gynecology, Neurology, Oncology, Orthopedics, Pediatrics).
    - **Clinical Audit Evaluation:** From an operational and clinical standpoint, having the same individual practitioners lead volume in unrelated medical specialties (e.g., Cardiology vs. Pediatrics or Oncology) indicates a severe data architecture flaw. This confirms that physician names were assigned semi-randomly or without relational constraints linking doctors exclusively to their accredited sub-specialties in the legacy system.
    - **Exclusion of Unassigned Categories:** `Unknown Department` and `Unknown Doctor` categories were excluded from the active breakdown to maintain executive visual clarity while documenting the underlying data governance gap.
- **Key Findings (Top Active Physicians per Specialty):**
    - **Cardiology:** Michael (37 visits), John (24), Christopher (22)
    - **Dermatology:** Michael (35), John (29), David (25)
    - **ENT:** Michael (42), John (26), James (25)
    - **Gynecology:** Michael (28), David (26), James (26)
    - **Neurology:** Michael (33), Christopher (24), David (22)
    - **Oncology:** John (31), Michael (30), David (30)
    - **Orthopedics:** Michael (35), Jennifer (26), Jessica (21)
    - **Pediatrics:** Michael (34), Jennifer (26), David (25)
- **Executive & Governance Recommendations:**
    - Enforce relational schema rules in the hospital EHR database to bind doctor profiles strictly to their primary clinical departments (Department_ID mapping).
    - Re-verify historical clinical provider assignment logs to separate true multi-specialty/consulting cases from data entry artifacts.

Section 8: Patient Demographics & Time Trends Results

8.1 Q8: Departmental Age Distribution & Data Quality Audit (Department_Age_Analysis)

- **Objective:** Determine the overall average age of patients and analyze age variation across specialized medical departments.

- **Data Summary:** Analyzed total patient visits across 9 medical departments and unassigned categories to evaluate central age tendencies.
- **Data Audit & Critical Quality Insight (Demographic Data Entry Flaw):**
    - **Severe Demographic Anomaly in Pediatrics:** The analysis reveals that the average patient age in the Pediatrics department (`Pediatrics`) is **48.2 years**. In a real-world clinical setting, a pediatric department caters exclusively to infants, children, and adolescents. An average age of 48.2 years represents an impossible clinical metric.
    - **Uniformity Across Unrelated Specialties:** Overall average patient age across all departments is tightly clustered around **47.3 years** (Cardiology: 47.1, Dermatology: 47.8, ENT: 47.4, Gynecology: 46.6, Neurology: 47.1, Oncology: 47.4, Orthopedics: 46.7, Unknown Department: 47.1).
    - **Analytical Diagnosis:** This near-identical average age distribution across unrelated departments, combined with the absurdly high average in Pediatrics, confirms that patient age values were populated semi-randomly in the legacy system without validation rules or age-range constraints relative to clinical departments.
- **Executive & Managerial Recommendations:**
    - Implement mandatory front-desk data validation rules (e.g., restricting Pediatrics patient age to <18 years upon registration).
    - Perform an audit on historical demographic logs to re-link patient ages with actual medical records.

8.2 Q9: Departmental Gender Distribution Analysis (Department_Gender_Distribution)

- **Objective:** Evaluate patient gender breakdown (Male, Female, Unknown) across specialized medical departments.
- **Data Summary:** Analyzed patient traffic across identified clinical specialties after filtering out the unassigned department category for executive cleanliness.
- **Key Findings:**
    - **Balanced Primary Gender Intake:** Patient volume is evenly distributed between Female (5,380 visits, ~41.4%) and Male (5,311 visits, ~40.9%) across the hospital.
    - **Departmental Gender Trends:**
        - **Orthopedics:** Highest Female volume with 691 visits compared to 622 Male visits.
        - **Oncology:** Led Female intake with 665 visits vs. 615 Male visits.

        - **Gynecology:** Recorded 674 Male visits (likely representing administrative/chaperone logs or system data entry noise) and 610 Female visits.
        - **Cardiology:** Slightly higher Male volume at 659 visits vs. 640 Female visits.
        - **ENT:** 639 Male visits vs. 590 Female visits.
    - **Data Audit & Quality Insight (High Rate of Missing Gender Logs):**
        - A significant volume of **2,309 visits (~17.8% of total hospital visits)** lacks a recorded gender (`Unknown Gender`).
        - These unrecorded gender entries are evenly distributed across all departments (~254 to 290 per specialty), revealing a systematic operational failure at front-desk intake where gender logging was routinely bypassed.
- **Executive & Managerial Recommendations:**
    - Mandate strict data entry checks at front-desk registration to ensure gender fields are populated prior to scheduling.
    - Perform targeted data enrichment on historical records with missing gender tags using primary patient profiles.

8.3 Q10: Temporal Visit Trend & Time-Series Analysis (Visit_Time_Series_Analysis)

- **Objective:** Evaluate the temporal trend of patient visits over months, quarters, and years to identify operational surges and seasonality.
- **Data Summary:** Analyzed all 13,000 visits across 2024 and 2025 broken down by quarters and months.
- **Key Findings:**
    - **Inter-Year Volume Stability:** Annual visit volume displays exact parity between 2024 (6,500 visits, 50.0%) and 2025 (6,500 visits, 50.0%), reflecting highly consistent systemic patient logging.
    - **Intra-Year Seasonality & Q4 Surge:**
        - **Q1 Baseline:** Starts at baseline volume (~506 to 558 visits per month; total Q1 = 1,600 visits in 2024 and 1,598 in 2025).
        - **Q2 to Q4 Escalation:** Traffic gradually builds through Q2 and Q3, reaching peak volume in **Q4** (1,698 visits in 2024; 1,636 visits in 2025).
- **Executive & Managerial Recommendations:**
    - Align clinical staffing, nursing shifts, and diagnostic facility capacity with Q4 peak surges.

    - Schedule preventative equipment maintenance and non-urgent administrative overhauls during Q1 to avoid operational bottlenecks during peak months.

Section 9: Advanced Analytics (Efficiency & Compliance Results)

9.1 Q11: Overall Follow-Up Time Gap Analysis (Overall_FollowUp_Gap_Analysis)

- **Objective:** Measure overall clinical scheduling efficiency and compliance by determining the average time gap (in days) between initial visit dates and scheduled follow-up dates across the hospital.
- **Data Implementation & Method:** Added a calculated column (`FollowUpGap`) in the primary dataset (`CleanedData`) using the formula `=[@FollowUpDate] - [@VisitDate]`, then aggregated the metric as an un-grouped global average (`Average`) in the PivotTable `Overall_FollowUp_Gap_Analysis`.
- **Key Findings:**
    - **Hospital-Wide Average Time Gap:** The overall mean duration between the initial visit and follow-up consultation is **32.98 days (~33 days)** across all recorded hospital visits.
    - **Operational & Patient Care Journey Insight:** This ~33-day baseline represents the standard post-consultation revisit interval for patient care management, serving as an operational benchmark for scheduling capacity and clinical compliance tracking.
- **Executive & Operational Recommendations:**
    - Establish a standard 30-day clinical revisit milestone across outpatient departments for chronic and post-operative monitoring.
    - Automate automated SMS/email follow-up reminders scheduled 3 to 5 days prior to the 33-day mark to minimize patient no-show rates.

9.2 Q12: Departmental Follow-Up Time Gap & Data Governance Audit (Department_FollowUp_Gap_Analysis)

- **Objective:** Evaluate variation in average follow-up time gaps (`FollowUpGap` in days) across specialized medical departments to detect operational or clinical scheduling disparities.
- **Data Summary:** Analyzed follow-up gaps across all 9 medical departments and the unassigned category.
- **Key Findings & Data Quality Audit:**

    - **Near-Perfect Uniformity Across Specialties:** Average follow-up duration is virtually identical across all clinical specialties, standing at exactly **33 days** for 8 departments (Cardiology, Dermatology, ENT, Gynecology, Neurology, Orthopedics, Pediatrics, Unknown Department) and **34 days** for Oncology.
    - **Data Architecture & Governance Gap:** In real-world healthcare settings, follow-up intervals vary significantly by medical field (e.g., Physical Therapy/Orthopedics require frequent weekly visits, whereas routine ENT or Dermatology consultations follow different protocols). The presence of a static ~33-day gap across totally unrelated medical specialties confirms that follow-up dates were populated semi-randomly or via fixed synthetic offsets (+33 days) in the legacy database.
- **Executive & Managerial Recommendations:**
    - Configure clinical scheduling protocols in the EHR system to establish department-specific revisit windows based on clinical urgency and specialty requirements.
    - Implement automated logic validation rules in database entry forms to prevent default/synthetic date assignments during follow-up registration.

9.3 Q13: Follow-Up Date Completion Rate & Data Governance Audit (FollowUp_Completion_Status)

- **Objective:** Evaluate patient care journey completion rates by determining the proportion of records with scheduled follow-up dates versus unassigned (Null/Blank) follow-up dates across all hospital visits.
- **Data Summary:** Analyzed 13,000 total hospital visit records across all departments.
- **Key Findings & Critical Governance Audit:**
    - **100% Completion Rate (Zero Blanks):** Exactly 13,000 visits (100.0%) feature a recorded follow-up date, with 0 records (0.0%) logged as Null, Blank, or None.
    - **Data Integrity & Architecture Diagnosis:** In real-world outpatient operations, a 100% follow-up scheduling rate is clinically impossible (as routine consultations, one-off diagnostic visits, and emergency triage rarely warrant mandatory follow-ups). Combined with the static ~33-day interval identified in Q12, this 100% completion rate conclusively proves that the legacy database employed an automated synthetic generator (`Synthetic Auto-Assignment`) that forcibly attached default follow-up dates to every record upon entry.
- **Executive & System Recommendations:**

    - Audit and re-engineer the database schema to allow optional/null follow-up fields for one-off clinical visits.
    - Mandate physician-driven follow-up scheduling in the EHR interface rather than automated system defaults.

Section 10: Advanced Analytics (Multivariate Cross-Analysis & Governance Rejection)

10.1 Q14 & Q15: Methodological Rejection & Data Integrity Audit (Multivariate_Age_Analysis)

- **Objective:** Evaluate potential multivariate cross-relationships between patient age, requested diagnoses/lab tests (Q14), and payment method correlations with clinical treatment procedures (Q15).
- **Executive Governance & Analytical Decision: Analysis Formally Suppressed & Omitted.**
- **Methodological Justification & Failure Analysis:**
    - **Garbage In, Garbage Out (GIGO) Principle:** As conclusively established in Section 8.1 (Q8 Departmental Age Analysis), the patient age attribute suffers from critical data corruption and synthetic randomization (evidenced by an average age of 48.2 years in Pediatrics and a uniform ~47.3 years average across all specialties).
    - **Rejection Rationale:** Constructing cross-tabulations or statistical correlations between diagnoses, lab tests, and a compromised demographic attribute (Age) yields invalid and misleading outputs. Proceeding with such analysis would violate analytical integrity and risk biasing executive strategic decisions.
- **Strategic Recommendations:**
    - Freeze all demographic-based clinical segmentation reporting until primary EHR database entry controls are remediated.
    - Re-audit and clean age/demographic master logs prior to performing future multivariate clinical modeling.